Revanth Palaparthi

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EDUCATION

Indian Institute of Technology, Dharwad — ***Dec 2021 – Apr 2025***
B.Tech in Computer Science and Engineering

Relevant courses: Artificial Intelligence, Software Systems & Laboratory, Data Structures & Algorithms, Data Base Management Systems, Design and Analysis of Algorithms, Operating Systems, Compilers, Deep Learning.

WORK EXPERIENCE

A Comparative Study of Deep Learning Architectures for Cervical Cell Segmentation in Pap Smear Images. (IEEE INCIT 2025, IIT Dharwad Research Publication):

- Conducted a comprehensive comparative study of eight deep learning models (U-Net, U-Net++, Attention U-Net, DeepLabv3+, PAN, SegFormer, YOLOv8, YOLOv11) for cervical cell segmentation on the SIPaKMeD medical image dataset.
- Engineered a modular PyTorch-based segmentation pipeline with patch-wise image cropping (512×512), advanced data augmentation (rotation, flipping, elastic transformation), and precise mask handling for improved boundary recognition.
- Achieved top performance using U-Net++ (Dice: 0.84, IoU: 0.78) and demonstrated competitive real-time inference with YOLOv11 (Dice: 0.83), highlighting trade-offs between accuracy and speed.
- Evaluated models using clinical-grade metrics (IoU, Dice, Precision, Recall, F1-score, Accuracy), supporting data-driven architecture selection for medical image analysis and diagnostic systems.

Defence Research and Developmental Laboratory (DRDL) (Summer intern):

Developed scalable, fault-tolerant network applications to optimize communication within a distributed system, utilizing packet retransmission strategies to ensure reliability and performance.

- Engineered real-time interaction protocols, enhancing session management and optimizing packet retransmission by up to 50%.
- Architected a simulation using TCP sockets for decentralized network management, resulting in a 50% increase in throughput.
- Deployed Ring Token Protocol to streamline communication and reduce latency across distributed nodes.

PROJECTS

Dynamic Social Interaction Platform:

- Engineered a basic social media platform with user authentication, posting, commenting, liking, and friending functionalities.
- Implemented server-side logic using Python, efficiently managing data transactions and server responses.
- Designed SQL schema and automated data population, streamlining database setup.
- Coordinated application flow through a modular Python architecture, enhancing user engagement and platform usability.

Algorithmic Trading System with Automated Portfolio Management:

- Developed an automated trading system integrating Zerodha Kite API, enabling real-time market data streaming, order execution, and trade monitoring.
- Implemented AI-driven trading strategies using technical indicators (RSI, MACD, Bollinger Bands) and historical data analysis, improving decision-making efficiency.
- Designed a backtesting module with Python and Pandas, enhancing strategy evaluation and optimizing investment performance.
- Engineered a secure authentication system with OAuth2 and multi-user access control, ensuring robust data protection and regulatory compliance.

Secure Online Voting System with Blockchain Integration:

- Engineered a web-based voting platform using the MERN stack (MongoDB, Express.js, React.js, Node.js), ensuring seamless user authentication and real-time vote updates.
- Integrated Ethereum blockchain smart contracts to enhance vote security, ensuring immutable, transparent, and tamper-proof election records.
- Developed a decentralized identity verification system using cryptographic hashing and smart contracts, preventing fraudulent voting and duplicate submissions.
- Optimized system performance with WebSockets for live vote tracking and implemented Web3.js for seamless blockchain interactions, enhancing transparency and accessibility.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java
Database Management: MySQL, MongoDB, Postgresql
Web-development: HTML, CSS, JavaScript, PHP, React, Node.js, Express.js, Tailwind
Scripting: Python, Bash
Libraries & Tools: NumPy, Pandas, Git, GitHub, Linux, Latex, Beamer, Flask

ACHIEVEMENTS

- **Awarded 3rd place in the Parsec 2024 Treasure Hunt**, showcasing strong problem-solving and teamwork skills in a highly competitive environment.
- Achieved **Led sponsorship acquisition for E-Summit '22 and '23**, securing over 10 industry leaders, resulting in increased event credibility and extensive networking opportunities for participants.
- **Ranked Top 5 in the 'DEVHACK' event at PARSEC**, IIT Dharwad's prestigious tech fest, by designing and developing an innovative web solution, engaging 100+ participants, and fostering collaboration among peers.
- Selected to attend the **9th International Conference on Information Technology (INCIT 2025)**, Thailand, organized by IEEE.

LANGUAGES

English, Hindi, Telugu